

Evaluation of the Multimeter's Bandpass

Students of the academic year 2024/2025

09/06/2025



UNIVERSITÀ
DELLA
CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

**MEASUREMENTS FOR TELECOMMUNICATION
SYSTEMS** by Prof. Francesco Lamonaca
A.A. 2024/2025

This work is part of the project:

EXTENDABLE

NETWORK OF **EXTENDED REALITY-ENABLED**
LABORATORIES FOR REMOTE PRACTICAL TRAINING

Web-site: <https://extendable.dieti.unina.it/>



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Italiadomani
PIANO NAZIONALE
DI RIPRESA E RESILIENZA

**The Research Team involving Units from:
University of Calabria, Naples Federico II, and Sannio
is proposing the creation of a network of laboratories**



Process

Goals

Instruments

Questions?

What are the instruments?

How to link them?



Procedure

Sinusoidal Signal

**Select a set of test
frequencies and record
the voltage**

Analysis

Data plot

Conclusion

Hard and soft skills



Goals

The purpose of this experiment is to evaluate the bandpass of the 34401A Digital Multimeter.

What are the instruments?

Multimeter: 34401A Digital



Capability:

- Measurement:

- 6½ digit resolution
- 10 measurement functions
- Basic accuracy: 0.0035% DC, 0.06% AC
- 1000 V max voltage input, 3 A max current input

- System

- 1000 readings/second
- 512 reading memory



What are the instruments?

Function Generator: Keysight 33220A



Key Features:

- 20 MHz Sine and Square waveforms
- Pulse, Ramp, Triangle, Noise, and DC waveforms
- 10 mVpp to 10 Vpp amplitude range
- AM, FM, PM, FSK, and PWM modulation types
- 14-bit, 50 MSa/s, 64 k-point arbitrary waveforms



How to link them?

One BNC-Banana coaxial cable



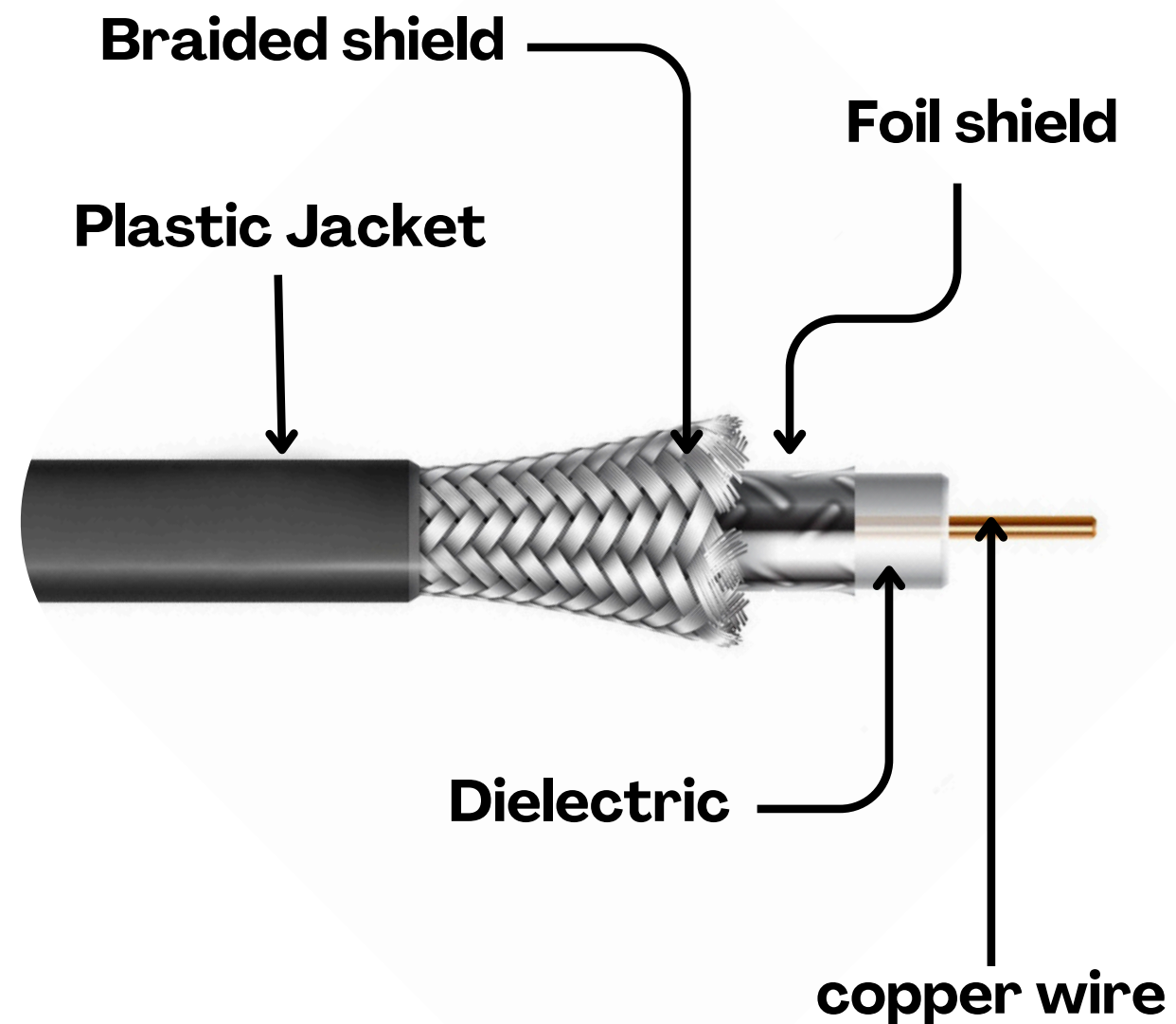
Why in parallel?

- Instruments share the same Voltage

Why a Coaxial Cable?

- Impedance closely matches the output impedance of the connected instruments
- Shield from external noise

How to link them?



Why the Plastic jacket?

- Shield from external interferences

Why the Braided shield?

- Prevent that the cable act as an antenna
- Protecting the signal from external noise
- Acting as a reference ground for the instruments.

Procedure

Sinusoidal Signal

f

1 kHz – 3 MHz

0 V

Offset

A

2 V_{pp}

**High
Impedance**

Out-Imp

Procedure

Select a set of test frequencies

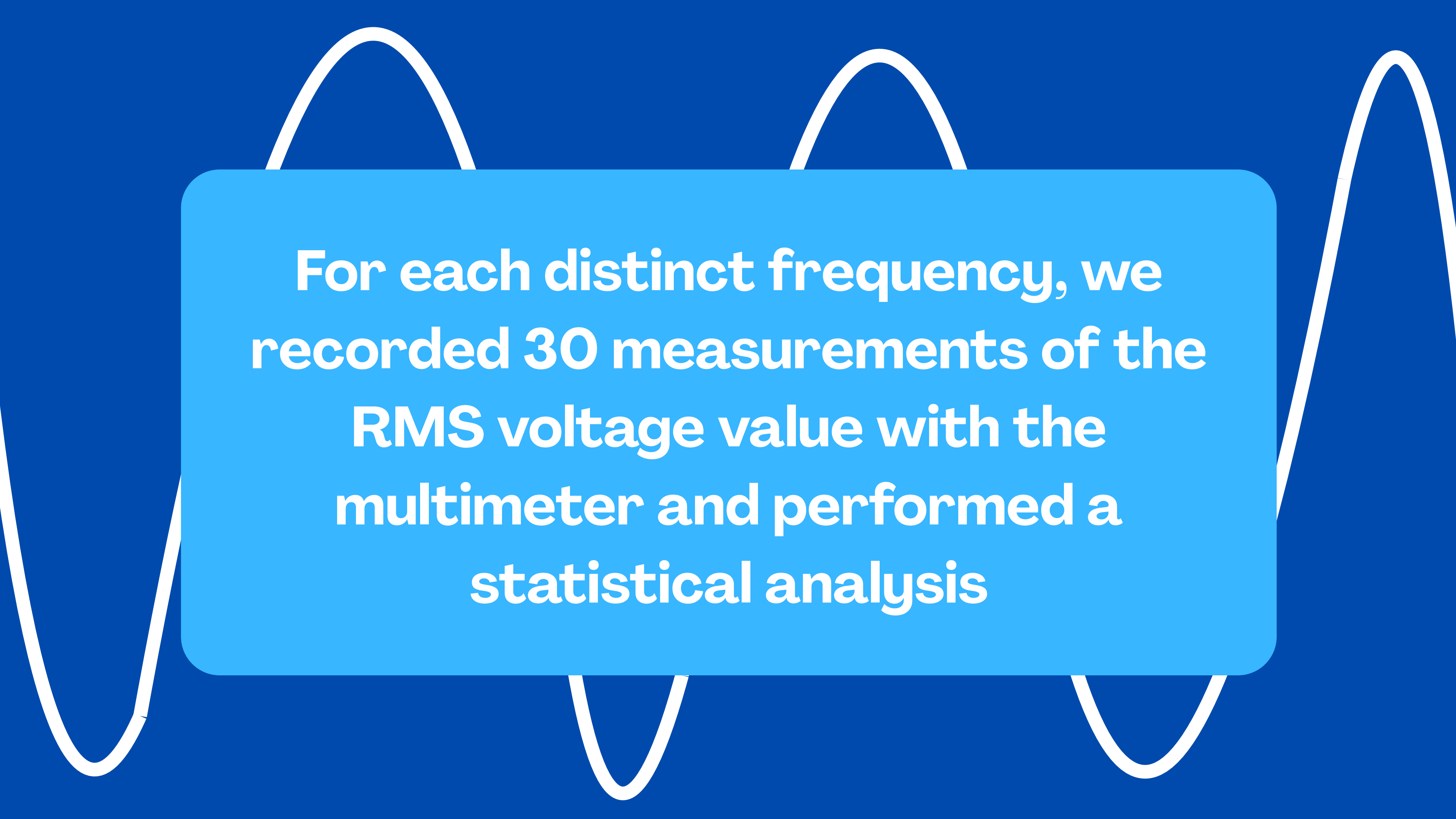
The set of frequency
from 1 kHz to 3 MHz

Step

- 1 kHz
- 50 kHz
- 150 kHz
- 250 kHz
- 300 kHz
- 500 kHz

and then

[1-3] MHz with
steps of 0.2 MHz



**For each distinct frequency, we
recorded 30 measurements of the
RMS voltage value with the
multimeter and performed a
statistical analysis**

Frequency	Mean Voltage (V)	Mean + Std Dev	Mean - Std Dev
1 kHz	0.70660	0.70666	0.70654
50 kHz	0.70626	0.70629	0.70623
150 kHz	0.70795	0.70798	0.70792
250 kHz	0.70746	0.70748	0.70743
300 kHz	0.70672	0.70673	0.70671
500 kHz	0.70735	0.70738	0.70732
1 MHz	0.71926	0.71928	0.71924
1.2 MHz	0.78487	0.78625	0.78349
1.4 MHz	0.91263	0.91285	0.91240
1.6 MHz	0.10948	0.10950	0.10946
1.8 MHz	0.11632	0.11642	0.11623
2 MHz	0.96988	0.97056	0.96921
2.2 MHz	0.74499	0.74516	0.74482
2.4 MHz	0.57900	0.57917	0.57883
2.6 MHz	0.45348	0.45356	0.45339
2.8 MHz	0.36268	0.36272	0.36264
3 MHz	0.29626	0.29629	0.29622

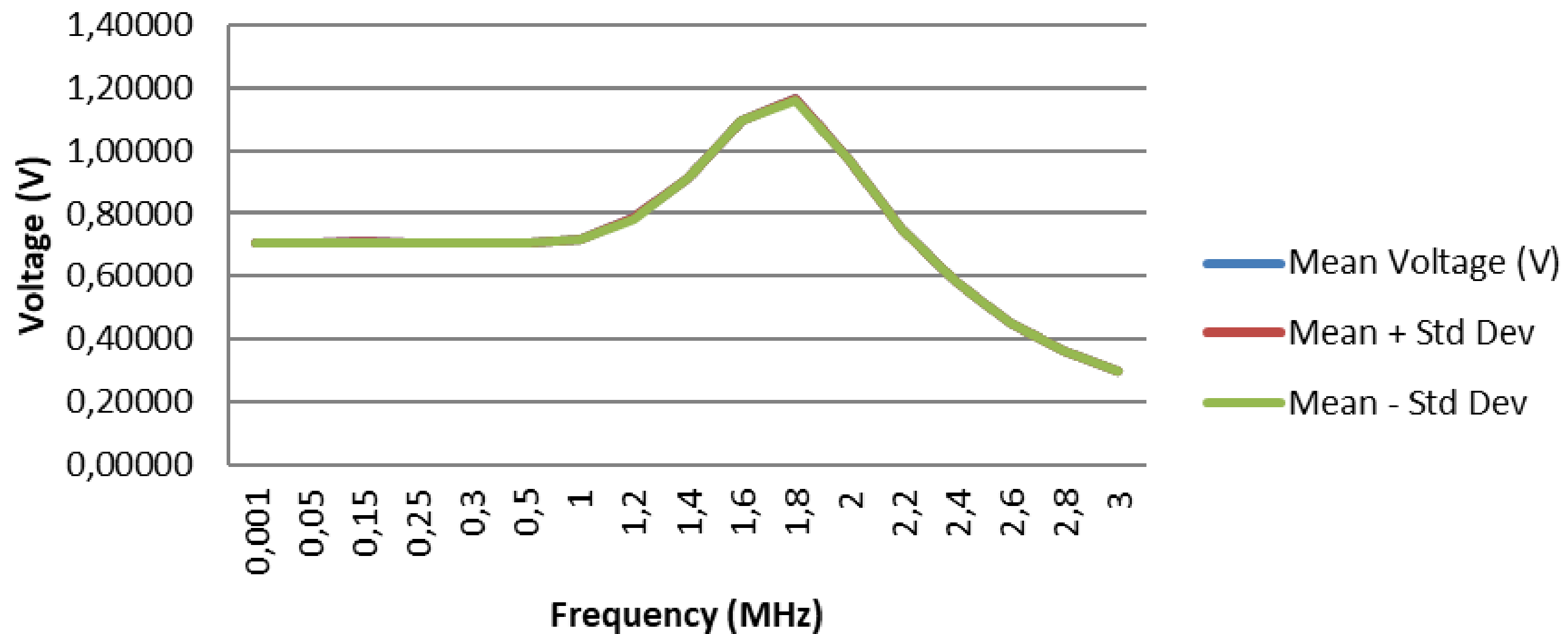
Analysis

**Mean and the standard deviation
are calculated on 30
measurements**

Analysis

Data plot

Combined Response



Conclusion

The device demonstrated:

- Accurate and stable readings between 100 and 300 kHz
- Reliable behavior up to approximately 500 kHz
- Outside this extended range, especially above 1 MHz, measurements became increasingly inaccurate, with deviations of up to 30%, in agreement with the technical documentation

Conclusion

Hard Skills Acquired

- **Instruments Set up**
- **Evaluation of the Multimeter's Bandpass**

Conclusion

Hard Skills Acquired

- **Data Analysis and Interpretation**
- **Multimeter and Function Generator usage**

Conclusion

Soft Skills Developed

- **Experiment Organization**
- **Teamwork and Collaboration**
- **Communication**

Conclusion

Soft Skills Developed

- **Problem-Solving**
- **Critical Thinking**
- **Time Management**
- **Attention to details**

Thank you for your attention

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